 DOI: *https://doij.org/10.10000/IJLMH.1110687*

INTERNATIONAL JOURNAL OF LAW MANAGEMENT & HUMANITIES

[ISSN 2581-5369]

Volume 8 | Issue 4

2025

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The Future Prospects of Drone Delivery Services (DDS) in the India: A Research Study

AJIT KUMAR DAS[1] AND DR. BHAGWAN SINGH[2]

ABSTRACT

Purpose: *The purpose of this research is to analyse the behavioural intention to use drone delivery services (DDS) in India, which is also known as Bharat. Specifically, the focus of the paper is on drone delivery services (DDS) in India and the ramifications associated with this phenomenon. This statement specifically highlights the immediate benefits of a particular project for India's economy. As a consequence, current research was conducted to investigate the method by which individuals in India are expected to utilise drone delivery services as a potential business possibility.*

Design/methodology/approach:

This study uses a purposive sampling method. Unified Theory of User Acceptance and Use of Technology has been used in this study (UTAUT2).We utilised the Partial Least Squares Structural Equation Modelling (PLS-SEM) technique to analyse an impressive 691 responses, confirming the robustness of our model. The goal of this study is to investigate how drone delivery services can facilitate entrepreneurs in identifying more business opportunities in the future. Furthermore, the general populace may gain a more comprehensive understanding of drone delivery services (DDS), which would empower them to make informed decisions about using DDS based on the knowledge they acquire.

Findings: The purpose of this study is to investigate the perceived risk associated with using drone delivery services, as well as the behavioural intentions to utilise these services. Within the framework of drone delivery services, this research examines the perceived risks and behavioural intentions related to drone delivery. We conduct all of our research with drone delivery services in mind. This study distinguishes itself from others in the IT sector by applying groundbreaking ideas to drone delivery services in India, an emerging market. Understanding the behavioural intention to use drone delivery services quickly is vital for firms and entrepreneurs who want to establish effective services.

Originality/ Value: *Indian DDS might be beneficial for people and the environment. Drones could help reduce carbon emissions and air pollution in cities by replacing delivery trucks*

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that rely on fossil fuels. DDS could create job opportunities for drone pilots, maintenance workers, and data analysts. The present study is the first time we've put numbers to these effects. To ensure that DDS is sustainable, compare its environmental benefits to those of traditional delivery methods. People's views and trust in UAVs have changed throughout time. Not India, but China, Rwanda, and the US have made progress in DDS. India can learn from the best DDS in the world. This study could facilitate the delivery of medicine carried by drones from Rwanda to India. The study may help India lead the way in drone logistics by identifying the most effective and efficient methods, as well as the most hazardous ones.

Digital India, made in India, and the Smart Cities Mission are all high-tech public service undertakings. Technology helps DDS improve their business and services. This article examines how DDS can be utilised in innovative ways to support national objectives. The Smart Cities Mission may improve city logistics, whereas DDS may enhance supply networks for rural agriculture. These kinds of synergies will demonstrate the importance of DDS to India's growth. Adopting DDS raises ethical and security issues, especially in India, which is very privacy-conscious and has many populations. The abuse of drones and the protection of data are both getting worse. This work could help India build drone delivery systems that are more moral. Discussions on drone technology cover issues such as data privacy, drone responsibility, and preventing illegal behaviour. Unlike other studies, this research utilises data to examine the growth and evolution of DDS in India. The study used predictive modelling and trend analysis to investigate sector-wide DDS adoption, identify difficulties, and Proposing solutions, predictions for the use of DDS in e-commerce, healthcare, and agriculture over the next decade may demonstrate its value. Such findings would enhance the scholarly significance of the study and inform policymakers and industry stakeholders.

***Keywords**: Unmanned Aerial Vehicle, last mile logistics, parcel delivery, drone technology Drone entrepreneurs*

I. INTRODUCTION

Drone delivery services (DDS), also known as unmanned aerial vehicle (UAV) delivery, have garnered significant attention from both the business and academic worlds for their effective use in package delivery. (Rao et al., 2016). This form of distribution is more flexible, easier to get to, faster and more efficient, safer, and even better for the environment than more traditional approaches. Because more people are interested in this technology, both academics and practitioners need to know a lot about how drone delivery works right now. Also, this technology doesn't make any toxic byproducts. As a result, scholars have shown a clear interest

in looking at the merits and cons of cargo drones. (Annaraud & Berezina, 2020). To meet customers' needs for high-quality services, save time, and maximise efficiency, e-commerce enterprises use the latest delivery technologies. Amazon, Google, and Walmart are just a few of the numerous online stores and e-commerce giants that are getting ready to offer drone delivery services. This approach is just one of many new ways for businesses to be creative. (Wu & Lin, 2018). When compared to other ways of delivering packages, drone delivery might have several benefits. These benefits can include shorter delivery times, lower maintenance expenses, and a method that is better for the environment. (Nadda et al., 2022). There are advantages and disadvantages for customers when it comes to using drones for carrying goods. Drone delivery has the potential to meet and surpass customers' needs when it comes to speed, flexibility, security, and cost. (Martin Joerss, 2016).

Drone delivery has the potential to be a more economical and eco-friendlier alternative to traditional delivery methods. Jeff Bezos, the CEO of Amazon, suggests that drones may be a more efficient alternative to trucks or people going to the store to deliver goods. Drone delivery services have gained prominence in the service industry recently. A few years ago, people were unsure about drone delivery services, but after successful tests in Iceland, Korea, and New Zealand, they are now eager for more.(Hwang, 2019). Modern consumers have increased expectations for innovative products, yet they often experience mixed emotions when considering the purchase and use of new technologies. This trend is attributed to a phenomenon known as "perceived risk," which causes people to be wary of trying new things due to the fear of potential negative consequences. "(Wu et al., 2012). The spread of drone delivery services is cast into doubt due to the paucity of research on the relationship between perceived danger and behavioural intention. This gap was filled by this research. Finding out what factors affect consumers' risk perceptions and their likelihood to employ drone delivery services was the driving force for this study. While introducing new technology-based services to the market, it is important to address client perspectives, particularly their concerns about the safety of drone delivery, as this study examines a novel delivery method and emphasizes

(A) Literature review and hypotheses development:

a. Drone delivery services:

(Dukkanci et al., 2024) This study examined the challenges of siting drone delivery facilities in proximity to other drone transport vehicles. Drones and cars visit these facilities. For hybrid vehicle-drone delivery, two vehicle types must be separated, unlike facility location. This analysis identified common concerns and suggested future investigation based on the extensive literature on installing drone delivery facilities. Aerial, ground, and hybrid vehicle-drone

delivery systems begin. It covers supply, drone launch and recovery, aerial or ground drone types, and discrete or continuous geographic locations. For each category, it analyses modelling methodologies, judgments, goal functions, limitations, and other aspects.

(Li et al., 2024) Online purchasing may benefit from drone last-mile delivery. To promote widespread adoption of drone delivery services, this study investigates how anthropomorphic features influence consumer acceptance. Theory uses anthropomorphism and planned behaviour. From Singapore, 450 replies were collected. Consumers' positive views of the human traits and interactions of delivery drones influence their adoption of drone delivery. The human features of delivery drones affect consumers' subjective norms, control, and attitudes.

(Dai et al., 2024) Drone delivery services are gaining popularity. The study analyses how perceived risk influences drone delivery. After conducting extensive literature research, the study categorizes aspects into five categories: human, drone, material, legal, and environmental. Key service demand factors can be analyzed using these categories. This study used C-OWA surveys and expert opinions to identify significant variables. The study found that flight safety, cargo security, and affordability are the primary drivers of demand for drone delivery services. Consumer perceptions of drone hazards matter, according to psychology. To mitigate drone delivery risks, this study can provide hands-on learning, demonstrate best practices, and offer valuable insights.

(Cheng et al., 2024) It has been discussed that more people are using drone delivery because it is faster and cheaper. The dispatcher must account for fluctuating wind conditions when scheduling drones from a depot for multiple destinations, which can delay deliveries. Methodology/Results: The delivery schedule is reliably optimised using our two-period drone scheduling approach to minimise wind uncertainty-related delays. By lunchtime, we update these timetables based on the current weather conditions. Our method reduces the essential riskiness index, which measures the likelihood and magnitude of lateness. The cluster-wise ambiguity set we employ to define uncertain flight times from wind observation data is easy to use and prevents overfitting the empirical distribution.

(Mishra & Tiwari, 2024) Faster, cheaper, and more eco-friendly urban deliveries could be the future thanks to unmanned aerial vehicles (UAVs), which are drones powered by batteries. Many businesses are initiating new initiatives to optimize drone logistics. However, drones' limited delivery range due to battery capacity is a big hurdle. With the proliferation of hybrid delivery systems combining car and drone technology, this issue may soon be a thing of the past. If conventional models keep assuming that drones can handle additional deliveries while

trucks park at client sites, parking constraints could reduce drone efficiency. Other approaches have been explored by researchers; however, these have their limitations, such as the need for massive batteries and blind spots and the use of charging stations installed on trucks. We propose resolving these issues by enhancing drone routing for quicker deliveries and designating particular locations to charge drones. Our MINLP (mixed integer non-linear programming) methodology can help you maximize transportation efficiency while investing in charging infrastructure. This method is effective, as shown by quantitative trials and sensitivity analysis, which aid in economic decision-making over a wide range of operating circumstances. After running the charging stations for three years instead of two, we were able to cut costs by 15%.

(Leon et al., 2023) Businesses are exploring the possibility of using drones for last-mile deliveries and may use them more frequently in the future. Privacy, regulations, trust within organizations, and practicality are some of the aspects that this study looks at as they pertain to the adoption of last-mile drone delivery services. A combination of structural equation modeling and partial least squares analysis is used to evaluate measurement models. The Amazon Mechanical Turk sample was based on data collected from an online poll. Perceived privacy risk is inversely related to drone delivery services' utilization, according to the research. Various factors, including one's privacy disposition, concerns about privacy, and legal protections, contribute to the perception of privacy danger. Customers' propensity to use delivery drones is mostly impacted by their perceived usefulness. The privacy of consumers is being safeguarded by the increasing practicality of drone delivery. Confidence in the business increases the likelihood that customers will use drone delivery services.

(Chen et al., 2022) Delivery by drone is something that the majority of people in impoverished countries have never heard of. It would be wise for retailers in developing economies to investigate the merits of last-mile drone delivery options. To better understand the personal and societal experiences of 391 individuals with drone delivery, this paper conducted an online poll. This work utilized Partial Least Squares Structural Equation Modelling (PLS-SEM) for its analysis. The study suggests that consumers' personal innovativeness and opinion passions influence their perceptions of ease of use. Users perceive drone delivery services unaffected by these two criteria. People were less interested in PEOU and drone delivery services because of privacy concerns. Unregulated, however, is the correlation between consumers' perceived utility of drone delivery services and the privacy risk they are willing to take in exchange for those services. Finally, there are theoretical and practical consequences for these results, as PEOU has a considerable positive impact on consumers' intentions to use drone delivery services, but

PU has little to no effect. On the other hand, we understand the factors that motivate people's privacy worries and their desire for last-mile drone delivery services.

(Benarbia & Kyamakya, 2021) In recent years, online retailers have witnessed an increase in both the number of daily package deliveries and the severity of customer expectations. As a result, parcel delivery became exceedingly costly, particularly for the last kilometre. To keep up with the competition and meet customer demand, businesses are seeking innovative autonomous last-mile delivery options, such as drones or autonomous unmanned aerial vehicles. Following their success in remote sensing and surveillance, drone delivery systems are now gaining traction as a novel approach to cutting delivery times and costs. A logistical necessity in the next few years will be autonomous drone sharing systems, thanks to the new rules and recommendations from the International Civil Aviation Organisation on how to organise these unique unmanned aircraft systems. This paper provides a thorough literature analysis and examines the performance levels of drone delivery systems by utilising a collection of pertinent research challenges, representative solutions, and concepts for developing and modelling logistics. In addition to outlining the key points, the report provides a concise overview of numerous potential research avenues for drone-based package delivery systems.

(D. Liu et al., 2020) Delivery by drone is something that the majority of people in impoverished countries have never heard of. It would be wise for retailers in developing economies to investigate the merits of last-mile drone delivery options. This paper conducted an online poll to better understand the personal and societal experiences of 391 individuals regarding drone delivery. This work utilized Partial Least Squares Structural Equation Modelling (PLS-SEM) for its analysis. The study suggests that consumers' personal innovativeness and opinion passions influence their perceptions of ease of use. Users perceive drone delivery services as unaffected by these two criteria. People were less interested in PEOU and drone delivery services because of privacy concerns. Unregulated, however, is the correlation between consumers' perceived utility of drone delivery services and the privacy risk they are willing to take in exchange for those services. Finally, there are theoretical and practical consequences for these results, as PEOU has a considerable positive impact on consumers' intentions to use drone delivery services, but PU has little to no effect. On the other hand, we understand the factors that motivate people's privacy worries and their desire for last-mile drone delivery services.(Shepherd, 2017) It should also be mentioned that the existing food delivery services, such those utilizing vehicles or motorcycles, have resulted in major issues, such as injuries, fatalities, and financial losses.

(Bamburry, 2015) From a supply-side perspective, utilising drone delivery services may result

in lower production costs. The reason being, drone delivery services are more efficient, using less fuel and fewer workers to accomplish the same task. In other words, companies that offer delivery services may experience several advantages. In addition to a potential drop in shipping costs, there was also an opportunity to ensure product delivery. UDS demand has surged in LMICs (poor and medium-income countries) due to e-commerce. Rapidly growing online shopping led to a greater demand for UDS in LMICs (Tamakloe et al., 2024) for delivery services to cities. The rapidity, security, and environmental friendliness of drones have aroused interest in new UDS modes. We looked at people's thoughts on drone delivery. A growing number of people are opting for drone delivery because to its low cost and ease of use. Dispatchers generate delivery delays by deploying drones to different locations from a central site while considering the speed and direction of the wind. (Cheng et al., 2024)

(Park et al., 2018) This study indicates that UAVs are gaining popularity for delivery because of their portability and ease of use in remote places. This study considers both urban and rural locations, examining the environmental impacts and anticipated improvements of drone and motorcycle delivery. The study also evaluated the environmental impact of electric motorbikes. Also taken into account were potential shifts in the country's approach to power generation. Researchers found that compared to motorcycle deliveries, drone deliveries produce half as many particles and a sixth of the global warming potential (GWP) per kilometre. When considering delivery distance, rural areas lowered environmental effects thirteen times more than urban places.

II. THEORETICAL BACKGROUND

(A) Technology Acceptance Model:

Theories have offered various frameworks for thinking about how to incorporate new technologies. One way is to act in a predetermined manner. The idea of perceived risk and its variants has received the greatest attention among these models. (Featherman & Pavlou, 2003a), the Technology Acceptance Model (TAM) (Venkatesh et al., 2003b), and the Theory of Planned Behaviour (TPB) (Ajzen, 2012) The following text is based on the theory of reasoned action. (TRA). According to TAM, the main factor that influences a person's intention to use a particular technology. is. There is a correlation between an individual's opinions of how useful and easy something is to use and their level of influence. Self-efficacy is one of the characteristics that researchers have attempted to incorporate into TAM in an effort to broaden its operational reach. Taylor and Todd (1995) and Lucas and Spitler (1999) conducted research to study the connection between users' views of social norms and their performance. Two of the

writers investigated this link. (Mathieson et al., 2001; Featherman and Pavlou, 2003) Among the factors that should be taken into consideration are both potential risks and oversights. Unified theory of acceptance and use of technology (UTAUT):

(Venkatesh et al., 2003b)

To create the UTAUT, eight key models and theories on technology adoption and usage were analyzed and integrated. Many theories and models exist about computer use, including the Motivational Model, the Technology Acceptance Model, the Theory of Planned Behavior, and combinations thereof. At UTAUT, we value hard work, positive social impact, and a positive work atmosphere. In addition to gender, age, experience, and voluntary usage, four other criteria moderate the association.

The original intent of the UTAUT model was to assess perceived usefulness and simplicity of use; it was created for the TAM study. Expectations of performance and effort are considered. In terms of actual technology utilization, the UTAUT model accounts for 40% and approximately 56% of the motivation to use technology. Since its introduction, this notion has been replicated and applied in numerous organizational contexts, improving its generalizability. The UTAUT is the product of an analysis and integration of variables from eight essential models and theories on technology acceptance and use. Computer use, the Motivational Model, the Technology Acceptance Model, the Theory of Planned Behaviour, and a combination of the two are just a few of the theories and models out there. At UTAUT, we value hard work, positive social impact, and a positive work atmosphere. In addition to gender, age, experience, and voluntary usage, four other criteria moderate the association. The UTAUT model, which aims to evaluate perceived utility and ease of use, was initially developed in the context of the TAM study. It takes performance and effort expectancy into account. About 56% of the motivation to use technology and 40% of actual usage can be explained by the UTAUT model. The generalizability of this concept has been improved through its replication and implementation in various organizational contexts since it was introduced.(Park et al., 2007)

(B) Unified theory of acceptance and use of technology UTAUT2.

(Venkatesh et al., 2003b) Afterwards, he presented and validated UTAUT2, enhanced the scope of the theoretical processes involved, and improved his UTAUT model. They incorporated routine, monetary worth, and pleasurable reward. We postulated that differences in age, experience, gender, and planning would reduce their impact on technology use and planned actions. In UTAUT2, compared to UTAUT, technology use increased from 40% to 52% and behavioral intention variance from 56% to 74%.

The term "performance expectancy" describes how much a person thinks a system will help them perform better on the job. This idea is comparable to relative benefit and perceived utility. The ease of use of our system is mirrored in the amount of work it requires. Concerning how difficult or easy it is to use; TAM is comparable to the diffusion of innovations hypothesis (DOI; Rogers 1983). Social influence, analogous to subjective norms in the Technology Acceptance Model (TAM), describes the degree to which a user considers the views of people around them when interacting with the system. "Facilitating conditions" refer to an individual's view of the overall organizational and technological framework of a system. This concept is similar to the TAM-TPB idea of perceived behavioral control. "Hedonic motivation" (HM) describes the positive emotional response one has when using a technological product or service. An app's price value (PV) is determined by users' mental weighing of the features' merits and cons against their expenses. One approach to describe habit is the degree to which people naturally carry out behaviors that they have learnt.

(C) Behavioural Intention:

There has been a lot of research on behavioral intention recently, and it's been included into several models on technology adoption. In contrast, a person's behavioural intention can be defined as the rationale for engaging in specific behaviors. Previous studies have shown that people tend to follow through on their declared intentions. Previous studies have shown that people tend to follow through on their declared intentions. Therefore, knowing the customer's goals might be enough to deduce their actions. (Ajzen, 2012)According to prior research, there is a correlation between one's stated intentions and their subsequent actions (Venkatesh et al., 2003b). According to prior research, there is a correlation between one's stated intentions and their subsequent actions. Consequently, it may be sufficient to understand the consumer's intentions to gain insight into their behaviour.

(D) Perceived risk theory:

"Perceived risk" is defined as "The anticipation of potential losses linked to a purchase serves as a deterrent to buyers. behaviour," as stated in the official definition of the term.." (Peter & Ryan, 1976). Perceived risk is "the kind and level of danger that a buyer feels when thinking about a particular purchase choice(Curran & Meuter, 2005). According to studies, customers perceive a behaviour as risky when they are unsure of the potential good or bad outcomes.(Poon & Tung, 2024).

It is more essential to describe risk in terms of the possibility for loss when it comes to explaining the risk that is linked with actions. The following is a discussion of the delivery

services that are provided by drones. When discussing the provision of drone delivery services, the phrase "perceived risk" refers to the individual's subjective assessment of the possibility of the drone being brought down. Previous research has laid the groundwork for this definition, which already existed. The concept of perceived risk explains how consumers assess prospective risks and strive to minimise the likelihood of unfavourable outcomes when making purchases.(Jensen, 2020).

1. Performance Risk:

(Hwang, 2019) Performance risk in drone delivery services refers to the possibility that the drone or its systems will malfunction and fail to complete the delivery. This includes the reliability, efficiency, and technical difficulties of the delivery service.

Drone delivery risk is crucial, as it impacts customers' enjoyment and trust in the firm. Technology reliability, regulatory compliance, and environmental factors that affect flight safety and delivery success all increase performance risk. Establishing rigorous testing processes, improving technological support, and developing innovative delivery strategies can reduce these risks and ensure reliable and consistent operations.

2. Delivery Risk:

When discussing the use of UAVs for package delivery, the term "delivery risk" describes the possible difficulties and unknowns that could arise. Any number of things could go wrong with the delivery, and that's what this risk is all about.

3. Privacy Risk:

a) **Data Collection:**

Drones equipped with sensors and cameras may inadvertently capture personal information about individuals and their residences while delivering goods. If intercepted or handled incorrectly, this information could serve espionage or other malicious purposes.

b) **Tracking and location sharing:**

To enable third parties to track drones, FAA regulations require that they broadcast their location using Remote ID. However, there is a risk of privacy breaches because individuals can be associated with specific deliveries, which reveals their habits and preferences.

c) **Consumer Trust:**

There is a problem with drone delivery and personal privacy. Unauthorized individuals may get access to services, which is a concern. Private information or keep tabs on people using data gathered by drones. Concerns about possible privacy invasions are understandable when it

comes to drones collecting visual data while flying, particularly in suburban and metropolitan areas. Take into account the essential aspects of privacy risk.

SLNo	Variables	Sources
1	Effort Expectancy	(Venkatesh et al., 2003a)
2	Performance Expectancy	(Venkatesh et al., 2003a)
3	Social Influence	(Venkatesh et al., 2003a)
4	Green Image	(Hwang et al., 2020) &(W. Kim & Hur, 2024a) (Chen et al., 2022)
5	Perceived Cost	(Venkatesh et al., 2003a)
6	Facilitating Condition	(Venkatesh et al., 2003a)
7	Perceived Risk	(Hwang et al., 2020) &(Yoo et al., 2018a)
8	Performance Risk	(Hwang et al., 2020) &(Choe et al., 2021) (Yoo et al., 2018b)
9	Delivery Risk	
10	Privacy Risk	

Sources: By Author

(E) Hypothesis Development:

According to Viswanath Venkatesh and Michael, a system's "Effort Expectancy" (E.E.) is the degree to which it is easy to operate. "Perceived ease of use (Technology Acceptance Model/TAM2) and complexity (Multidimensional Perceived Control of Use). (Thompson et al., 1991), and ease of use (IDT) (Gary C. Moore and Izak Benbasat, 2016)" serve as the basis for E.E. Academics from all over the world have studied the correlation between E.E. and B.I. in connection to the use of drone delivery services. E.E. and B.I. were favourably and strongly connected with the use of drone delivery services, according to the data. A couple of instances are Hewetal (2015), Peng et al. (2018), and Tam et al. (2018). An individual's inclination to use drone delivery services can be greatly impacted by factors like perceived complexity, convenience of usage, and usefulness. In a similar vein, more relevant factors might be included. Consequently, the research can test the following hypothesis:

H1: There is a positive and significant relationship between EE and BI to use drone delivery services.

1. Performance Expectancy (PE)

(Venkatesh et al., 2003b) People believe this is PE when they are confident that technological advancements will benefit workers' jobs. According to (Martins et al.2014), behavioural intention is inconceivable without PE. Users of PE are also more likely to believe that drone delivery services will simplify their work.

H2: The use of drone delivery services is associated with a positive and substantial link between PE and BI.

2. **Social Influence**

What we call "social influence" (SI) are the external factors that exert pressure on even the most technologically sophisticated societies. One example of an outside force that could impact a customer's choice to use a service is social influence. (Shih & Fang, 2004).

H3: There is a positive and significant relationship between Social Influence BI to use drone delivery services.

3. **Facilitating Condition:**

When using targeted technologies, it helps if people believe that help is always available in the form of infrastructure and support. This viewpoint makes it possible to employ focused technological approaches. (Venkatesh et al., 2003b),

H4: Facilitating condition has a positive significant relationship on BI to use drone delivery services.

4. **Green Image:**

A company's "green image" is the impression it gives to customers when they think of it in terms of its dedication to and care for the environment. (Mathew et al., 2021)

H5: Green Image has a strong favourable effect on BI to utilize drone delivery services.

5. **Perceived Cost:**

Perceived costs significantly influence consumer behaviour since individuals typically evaluate these aspects in comparison to the benefits they anticipate receiving from the product or service. For example, a person might view a low-cost product that demands considerable time and effort to use as being more expensive than a higher-priced product that is easy to use and requires less time.

H6: Perceived cost has a positive significant effect on BI to use drone delivery service.

6. Perceived Risk:

In e-healthcare research, researchers have actively investigated perceived risk in conjunction with UTAUT, which exposes several consumer risks that may influence consumers' adoption of BI and their use of products and services (Ben Arfi et al., 2021)

H7: Perceived risk has a positive significant relationship on BI to use drone delivery services.

7. Performance Risk:

Due to poor execution, a project, investment, or commercial activity may fail to meet its goals. Technical issues, resource constraints, poor management, or unexpected events could be the cause. Performance risk is especially high in projects or investments that depend on integrating multiple components or executing complex operations.

H8: Performance risk has a positive significant relationship on BI to use drone delivery services.

8. Delivery Risk:

The failure to complete a project, product, or service on schedule, within budget, or to quality requirements can have significant consequences. It is crucial in project management, supply chain management, and service delivery because missed delivery deadlines can cost money, reputation, and customer satisfaction.

H9: Delivery risk has a positive significant relationship on BI to use drone delivery services.

9. Privacy Risk:

The risks associated with disclosing, misusing, or improperly managing personally identifiable information include identity theft, financial loss, reputational damage, and a loss of independence. Today's increasingly digital world, with its rise in electronic data storage and transmission, presents all the risk associated with collecting, storing, processing, exchanging, and destroying personal data.

H10: Privacy risk has a positive significant relationship on BI to use drone delivery services.

(F) Behavioural Intention:

An individual's preparedness or intention to act is a crucial element. It supports numerous psychological and behavioural ideas, particularly in social psychology and technology adoption. The probability of an activity's completion escalates with the intensity of its purpose.

(G)Research Methodology.

A 5-point Likert scale survey was used in the exploratory study. The middle is Neutral, and the scale goes from Strongly Agree to Disagree. Khraim (2021), Viswanath Venkatesh, and Michael G. Morris (2003) were consulted for the research products and questions, which were then adjusted to suit the needs of this study. Through the use of both online and offline surveys, this study gathered data from various states in India. A convenience sample of ten people served as a pilot group to whom the preliminary questionnaire was sent. The validity of the questionnaire's contents was guaranteed by this measure. This pilot group consisted of academics and management experts. To make the questionnaire more accurate, feedback from the pilot group was incorporated into it. The study used both online and offline techniques to disseminate 350 surveys. Unfortunately, 195 out of the 218 questionnaires that were received were not suitable for the study because they were either incomplete or contained inaccurate information. To explore the nature of the relationship among the variables evaluated, the study employed the PLS-SEM technique. Collected the questionnaire from a variety of individuals, business owners, and young people, primarily students, in the Indian states located in various regions of the country through personal visits and online methods. To ensure the sample represented the entire population, it included individuals from various backgrounds. This encompassed people of both genders, all educational levels, various occupations (including both self-employed individuals and those employed by others), members of the service class, and residents from both urban and rural areas.

(H) Demographic Profile:

The basic info about the people that filled out the survey in **Table 1**. Of the total 691 responders, 304 are men and 387 are women, making up 44% and 56%, respectively. Of those who took the survey, 23.5% were in the 18–24 age bracket, 39.2% were in the 25–34 age bracket, 20% were in the 35–44 age bracket, 10% were in the 45–54 age bracket, and 7.3% were in the 55+ age bracket.

Frequency Distribution for Respondent's Demographics: Table 01

Demographic Characteristics N=691	Responses	Valid Percentage
Gender		
Male	304	44%
Female	387	56%

Age

1. 18-24	162	23.5%
2. 25-34	270	39.2%
3. 35-44	138	20%
4. 45-54	69	10%
5. 55 & above	52	7.3%

Educational Qualification

1. 10+2(Higher Secondary)	28	4%
2. Bachelors (BA. BCOM, BBA, BSC etc.)	256	37%
3. Masters (MA, M.COM, MSC, M-TECH&MBA)	345	50%
4. Any Technical Degree (Including ITI, Diploma etc.)	21	3%
5. Doctorate	41	6%

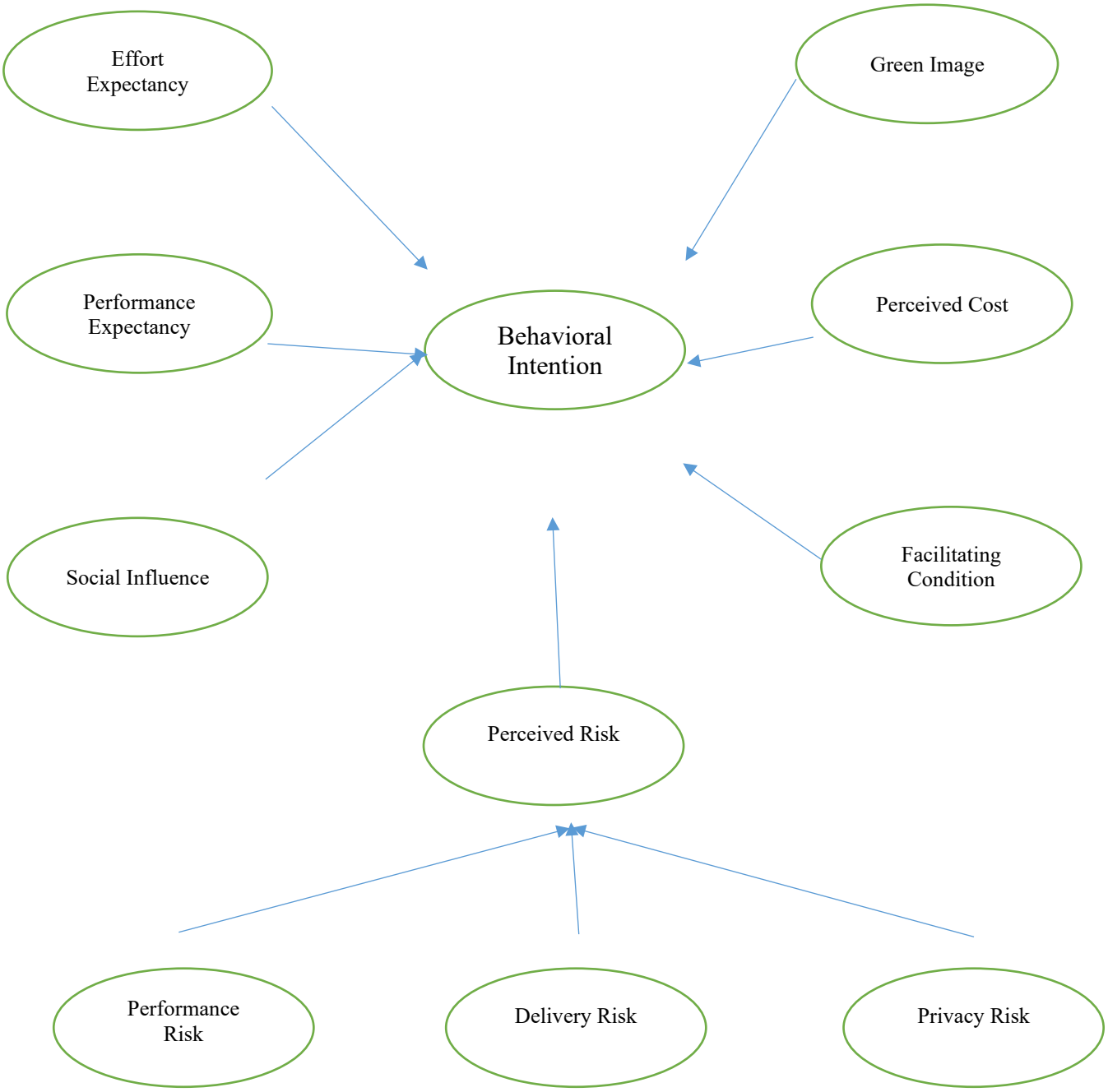
Status of respondents As

1. Students	415	60%
2. Government / PSU Employees	34	5%
3. Private Job	145	21%
4. Self Employed	35	5%
5. Business Persons	62	9%

Measurement of Scale:

To determine the scale's reliability, the study used Cronbach's alpha and composite dependability (Munir, 2018). The permissible range for Cronbach's Alpha in this study is 0.762 to 0.875, which is greater than the limit of 0.6 (Fornell & Larcker, 1981). The range of Cronbach's alpha values was examined in this study.

Figure:1 (Relationship Tested)



Sources: By Author

Table-2: "Reliability and Validity of the scale"

	Construct's	"Cronbach's Alpha"	"Rho_A"	"Composite Reliability"	"Average Variance

					Extracted
1	BEHAVIOURIAL INTENTION (BI)	0.839	0.841	0.892	0.675
2	DR	0.855	0.859	0.896	0.633
3	EE	0.8550	0.858	0.896	0.632
4	FC	0.857	0.858	0.893	0.583
5	GI	0.881	0.882	0.913	0.678
6	PC	0.857	0.859	0.891	0.538
7	PE	0.845	0.847	0.889	0.617
8	PER RIS	0.871	0.872	0.903	0.609
9	PFR	0.746	0.747	0.855	0.663
10	PRIR	0.856	0.856	0.896	0.634
11	SI	0.891	0.892	0.911	0.5344

Sources: By Author

III. Result and analysis

Using the UTAUT 2 model and the element of perceived risk that increases behavioral intention, the study seeks to uncover the factors that have an impact. Potential drone delivery services in India are the subject of this study. The factors' established and proven correlation is the focus of this investigation.

Relationship Formation: The interrelationships of the variables under investigation are shown in Figure 2. In order to determine if the suggested idea is appropriate, the research does a validity test.

Discriminant Validity: This measures the variety of constructions. An indication of discriminant validity is a low level of collinearity between many constructs. Words like "HTMT," "cross loading," and "Fornell-Larcker criteria" can describe its quantitative nature.

HTMT: Consideration of the HTMT value is necessary for drawing the conclusion that the contracts utilized in the study were discriminatory. A minimal discriminant validity score of

0.85 or lower indicates that the HTMT is functioning as intended. Any number between 0.508 and 0.828 in Table 2 is considered to be within the acceptable range. (Somjai et al., 2019)

Fornell- Lacker Criterion:

Thirty years ago, the FLC was established to assess the concept's appropriateness in terms of discriminant validity. The diagonal is said to be stronger if, by the "Fornell-Larcker criterion," its initial value is greater than the values.

Figure :2 Path Analysis

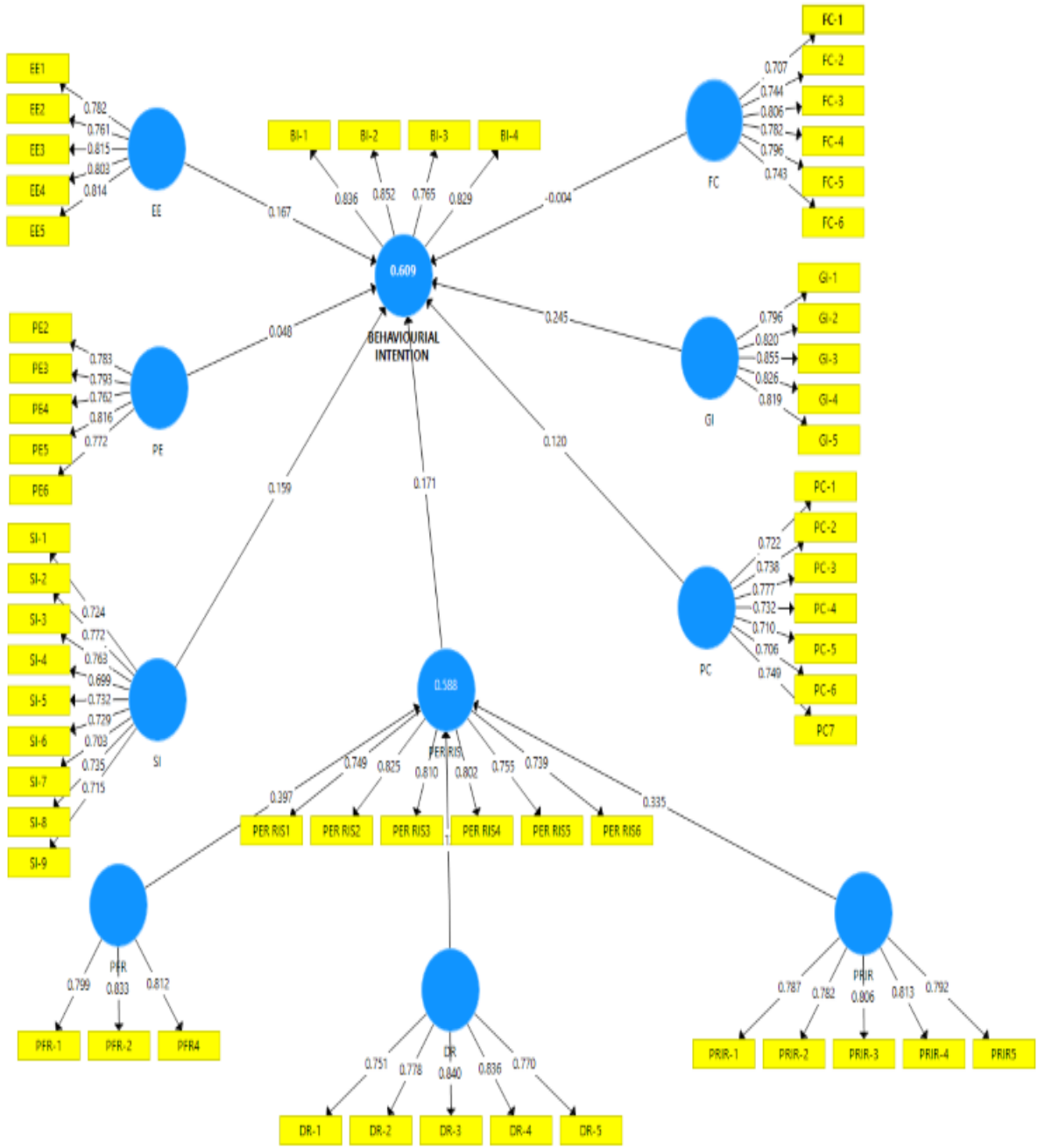


Table:3

	BEHAVI-OURIAL INTENTION	DR	EE	FC	GI	PC	PE	PER RIS	PFR	PRIR	SI
BEHAV-IOURIAL INTENTION	0.821										
DR	0.593	0.796									
EE	0.647	0.542	0.795								
FC	0.647	0.547	0.888	0.764							
GI	0.690	0.594	0.631	0.663	0.823						
PC	0.666	0.635	0.634	0.645	0.695	0.734					
PE	0.617	0.520	0.643	0.656	0.663	0.633	0.785				
PER RIS	0.669	0.637	0.640	0.668	0.658	0.740	0.634	0.781			
PFR	0.607	0.639	0.527	0.506	0.567	0.635	0.529	0.691	0.814		
PRIR	0.674	0.743	0.568	0.584	0.618	0.672	0.560	0.681	0.620	0.796	
SI	0.696	0.589	0.707	0.741	0.726	0.726	0.737	0.714	0.612	0.659	0.730

Sources: By Author

Cross loadings: As stated by(Hair et al., 2019) It is a way of describing structures that undergo cumulatively heavy loadings. In a typical column, the value in the diagonal cell should be bigger than all the other values for discriminant validity to be considered adequate. Assuming the same column is affected. Numbers that are diagonally adjacent in a certain column are significantly smaller than the normal-sized numbers. Consequently, research on discriminant validity is fraught with difficulty. (Kock, 2015).

VIF: According to Hair Jr. et al. (2017It's a sign of strong or multiple collinearities between separate mental operations. Variance inflation factor (VIF) values between 1.859 and 2.454 fall below the threshold of 3.3, making the planned study a good fit for factor-based PLS SCM. (Kock, 2015)

R2: This exemplifies how the exogenous factors can be used to provide an explanation for the endogenous variables within the context of the research investigation. The following are the three different values for R2 that have been identified by Hair et al. (2019): a big value of 0.75, a moderate value of 0.50, and a very small value of 0.25. As an illustration, the value of BI is 0.627, which is a desirable figure due to the fact that it is higher than 0.25 and, as a result, represents a reasonable degree of R2.

Table:7

	R Square	**R Square Adjusted**
BEHAVIOURIAL INTENTION	0.629	0.626
PER RIS	0.605	0.603

F2: The presence of higher (f2) values is indicative of the fact that independent constructions exert a greater influence. Cohen (1988) provides a definition of (0.02). In terms of a medium impact, 0.15 represents a minor impact. The value is 0.35 times higher than the influence. Whereas in the variables examined in the research study have the range of 0.003 to 0.067 did not significantly affect the dependent variable.

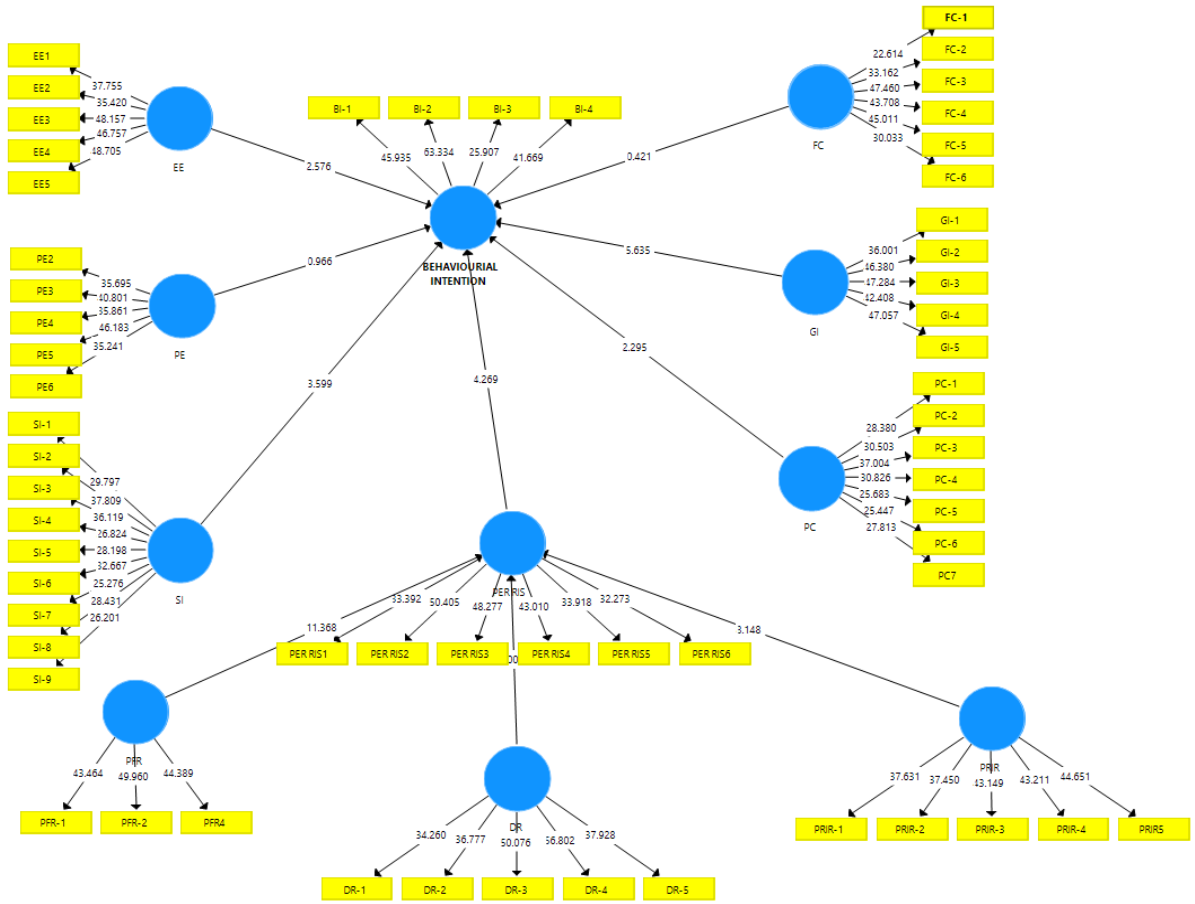
	BEHAVI-OURIAL INTENTION	**DR**	**EE**	**FC**	**GI**	**PC**	**PE**	**PER RIS**	**PFR**	**PRIR**	**SI**
BEHAV-IOURIAL INTENTION											
DR								**0.017**			
EE	**0.014**										
FC	**0.000**										
GI	0.055										
PC	**0.011**										

PE	**0.002**										
PER RIS	0.035										
PFR								**0.212**			
PRIR								0.120			
SI	0.027										

Sources: By Authors

The second phase involves testing the hypothesis through the bootstrapping technique in PLS-SEM. In this step, we test the hypothesis. To get estimated "T-values" for figuring out how important structural path measurements are and "P-values," bootstrapping needs to take a lot of subsamples from the main sample and replace them with new ones. This gives the bootstrapping standard error.

Figure:3 (Bootstrapping)



Sources: By Author

Table:9

Hypothesis	Constructs	Original Sample (O)	Sample Mean (M)	Standard Deviation (STDEV)	T Statistics (|O/STDEV|)	C.I @2.5 %	C.I@97.5	P Values	Decision
H9	**DR -> PER RIS**	0.133	0.134	0.044	3.003	0.045	0.045	**0.003**	**Accepted**
H1	**EE -> BI**	0.167	0.166	0.065	2.576	0.037	0.037	**0.010**	**Accepted**
H4	**FC -> BI**	-0.030	-0.030	0.070	0.421	-0.167	-0.167	**0.674**	**Rejected**
H5	**GI -> BI**	0.235	0.235	0.042	5.635	0.153	0.153	**0.000**	**Accepted**
H6	**PC -> BI**	0.110	0.111	0.048	2.295	0.016	0.016	**0.022**	**Accepted**
H2	**PE -> BI**	0.039	0.040	0.040	0.966	-0.042	-0.042	**0.334**	**Rejected**
H7	**PER RIS -> BI**	0.193	0.192	0.045	4.269	0.104	0.104	**0.000**	**Accepted**
H8	**PFR -> PER RIS**	0.397	0.396	0.035	11.368	0.329	0.329	**0.000**	**Accepted**
H10	**PRIR -> PER RIS**	0.345	0.345	0.042	8.148	0.261	0.261	**0.000**	**Accepted**
H3	**SI -> BI**	0.198	0.198	0.055	3.599	0.087	0.087	**0.000**	**Accepted**

Sources: By Author

IV. Discussion and Recommendation

Drone delivery is gaining popularity in the business world, and academics have recently focused on the big picture, encompassing efficiency, adaptability, safety, and cost. (*2016 the future of last mile*, n.d.) People could choose to physically and socially isolate themselves during the COVID-19 pandemic by using delivery drones. Although there were benefits to using vehicles like cars, trucks, and motorcycles, such as faster delivery and a lower environmental impact

compared to more traditional methods, they were nevertheless underutilised. (Rose, 2013a) Results from both functional and non-functional analyses reveal a strong correlation between user behaviour in DDS and the behavioural goals they've established for themselves. Furthermore, the findings suggest that social influence is the factor that has the most influence on the intention to engage in a particular behaviour. with other people's influences following closely behind Price sensitivity, ease of use, and enabling conditions influence customer behavior. However, the potential privacy and performance concerns of DDS significantly hinder widespread adoption of the technology. Various factors are considered, including the following: social impact, perceived cost, perceived risk, performance risk, delivery risk, privacy risk, and effort expectancy. Then there's the green image and perceived risk.

The first hypothesis: **Effort Expectancy has a positive effect on the use Behaviour of Drone Delivery Services (DDS) by customers.**

Consequently, the findings of the regression analysis strongly supported the hypothesis that behaviour is a critical factor in the onset of DDS. These results from prior studies support the overall conclusion stated.

The Second Hypothesis Despite predictions, there was no discernible positive influence when looking into behavioural intention to use DDS's services. Clients are more likely to utilise DDS's services if they perceive them as beneficial and believe they can improve their work processes. (Hwang et al., 2020:,Kim & Hur, 2024)

The Third Hypothesis discusses the positive effects of stress and social influence on motivation and behavior, supported by regression analysis. It emphasizes the role of support from friends, family, and coworkers in enhancing motivation, particularly in the context of technology adoption. The text highlights the positive effects of stress and social influence on motivation and behavior, using regression analysis for support. It emphasizes the importance of support from friends, family, and coworkers in enhancing motivation, especially regarding technology adoption. Furthermore, it points out that early adoption can have a significant impact on society, although continued use may lead to devaluation over time. .(Gefen et al., 2003)

The Fourth Hypothesis:

When it comes to DDS, the facilitating circumstance isn't doing anything beneficial. The accessibility of resources is a significant factor in determining whether users will utilise DDD services. Here, we review the abilities, facilities, prior knowledge, and peer assistance that comprise these resources. Previous research has highlighted the importance of new technology installations for facilitating consumers' behavioural objectives.

The Fifth Hypothesis: The perception of drone delivery services as environmentally benign positively influences the behavioural intention to use DDS. It found that people's views on using drone delivery for food are greatly influenced by their impression of these services as being ecologically beneficial. Since customers are more inclined to see drone delivery services favourably if they perceive them as environmentally conscious, it stands to reason that this bodes well for the services. (Hwang & Kim, 2019)

The sixth hypothesis: DDS may be a cheaper and less environmentally damaging option than other alternatives, according to the study's sixth hypothesis. Jeff Bezos, the head of Amazon, is of the opinion that drones could provide a more effective means of delivering products compared to trucks or in-person visits. The analysis focused on carbon dioxide emissions, integrating variables such as energy consumption, delivery frequency, and the size of service zones in the comparison. Drone delivery had higher energy requirements than vehicle delivery, but lower carbon emissions, and fewer delivery stops as the service zone approached the depot(Rose, 2013b).

The Seventh hypothesis: According to the seventh hypothesis, the positive views of risk that customers have have an effect on their behavioral intentions to accept DDS products. A common practice is to incorporate perceived risk as an additional component into both the TAM and the DOI measurements**.** In a 2007 study, Herzenstein et al. found that consumers may hesitate to purchase products and services that utilise cutting-edge technology because they are often not well-informed, leading to feelings of fear or mistrust. "Featherman and Pavlou" (2003b) describe the ", Possibility of losing something while trying to achieve a goal by using an e-service" (p. 454) as the dangers of using an online service. The degree of risk linked with different goods and services can vary, and people's perceptions of risk are complex, according to research.

The eighth hypothesis is the current study shows that a performance risk positively impacts customers' behavioural intention to use DDS. The purchase of DDS is guaranteed if it performs well.

The ninth hypothesis: There was a positive link between the levels of risk associated with delivery and the findings of the study. The possibility that their package will not arrive due to unanticipated circumstances, such as theft or damage, is a concern that is frequently considered by individuals. It is important to take into consideration this concern as a sign of delivery risk, which is the likelihood that a transaction could result in delivery problems

The study benefits from the potential threat to an individual's privacy. One definition of "privacy

risk" is the possibility that someone may use your personal information in a way that you are unaware of or have not given them permission to do (Featherman & Pavlou, 2003, page 455)

Factors	Recommendations
Effort Expectancy (EE)	Drone delivery is relatively new, particularly in emerging areas where consumers have limited exposure to it. Nguyen et al. (2023) studied the introduction of drone delivery services in an emerging country and found that perceived ease of use (effort expectancy) predicts consumer acceptability. In markets unfamiliar with drones, users are more likely to adopt the service if they see it as simple.
Performance Expectancy (PE)	The study examines how consumers' decision to use drone delivery services is influenced by their performance expectations. According to the study, consumers who think drone delivery would improve their service experience by reducing wait times and providing more precise delivery windows are more likely to use it. Perceived performance improvements such as convenience and efficiency closely correlate with consumer readiness to transition to drone delivery.
Social Influence (SI)	For the purpose of raising people's knowledge of any new technology, it is strongly recommended that marketing campaigns be carried out through a variety platform of the media, including television, newspapers, and the internet, among others. during the early stages of adoption to increase people's awareness of this technology and ensure the success of word-of-mouth marketing. There will be a significant influence on individuals as a result of the quick increase in awareness of the use of DDS. By exerting influence over one another, they are making use of this emerging technology. This will significantly impact adoption in the long run.
Green Image (GI)	Businesses are using their green image to attract eco-conscious customers as environmental sustainability becomes more important. The "green" image of drone delivery services has the potential to increase acceptance among consumers by portraying drones as a more environmentally friendly alternative to conventional delivery methods.

Perceived Cost (PC)	Decisions about implementing drone delivery services are affected by the perceived costs for businesses and customers. Factors influencing this technology's adoption include initial investment, operational costs, and potential long-term savings.
Facilitating Condition (FC)	Businesses need training and technical support to use drone technology. Companies must teach staff to operate, maintain, and integrate drones into logistics. These conditions minimize entrance barriers, making the technology more accessible to organizations of all sizes. Smartphone availability, excellent internet access, and awareness campaigns can help consumers adopt drone deliveries. Drone delivery companies are more popular in locations where consumers feel supported in using the new technology.
(PERCEIVED RISK (PER) 1. Performance Risk (PR) 2. Delivery Risk (DR) 3. Privacy Risk (PR)	Risk perception can slow or speed up drone delivery service uptake by businesses and customers. Transparency, solid security protocols, and consistent service performance help minimize privacy and operational issues, making drone delivery more appealing. Consumers and organizations perceive risk as the negative repercussions of adopting a new technology. Businesses and customers' willingness to use drone delivery services depends on perceived risk. Privacy and safety concerns may lead to corporate operational hazards.

Sources: By Author

(A) Benefits to the CAGS (Corporate, Academia, Government and Society).

1. Corporate: Logistics cost reductions benefit companies considerably. Drones eliminate the need for drivers, vehicles, and gasoline, saving money. By avoiding gridlock and using direct flights, organisations may meet same-day and same-hour delivery needs. Amazon and Walmart improve supply chain efficiency and consumer satisfaction with drones. Many companies prioritize sustainability for customers and the environment. Electric drones reduce carbon emissions in greener logistics. Such an approach decreases the company's carbon footprint and boosts its eco-friendly brand image. A corporation can showcase its IT skills with drone delivery. Representing early adopters as innovative and customer-focused fosters loyalty and draws new customers.

2. Academia: To work as a drone delivery service, you need to know a lot about logistics, tech,

and computer science. Universities with programs that study drone technology can meet this need. Universities could improve flying patterns, use AI to help drones find their way on their own, or make drones that can fly in all kinds of weather. Drone technology improves STEM subjects. You can learn by doing in classes like UAV programming, AI, and aerodynamics. Business schools can also use case studies with drone transportation to show how they can use innovation and supply chain management. Robotics, data analytics, and UAV repair jobs are growing because of drones. Corporations and schools can work together to teach kids about new jobs. These deals get graduates ready for what employers want, making the job market more trained and competitive.

3. Government: Drones can transport vaccines and blood to remote or disaster-affected regions. They can go around damaged roads or buildings to provide supplies in an emergency. Rwanda has successfully used drones to bring medical supplies to people in rural areas. Investing in drone ports and air traffic management systems can help the economy expand by getting more people to use drone technology. Jobs in manufacturing, software development, and following the rules help the local economy. Using drones to deliver things helps efforts throughout the world to cut down on carbon emissions. Countries can use drone delivery systems to combat climate change and be more eco-friendly.

4. Society: Drones are particularly helpful during emergencies, as they can quickly deliver food, water, and medical help when disasters like floods or earthquakes strike. Being able to navigate challenging terrain helps humanitarian groups save lives and accomplish tasks more efficiently. To cut down on traffic and pollution in cities, use drones instead of delivery vehicles that run on gas. This change enables cities to meet their health and environmental objectives. Drones could make the world a better place by making things easier and more accessible. We need to address issues with rules, technology, and culture to achieve these benefits. Drone delivery systems can enable people to collaborate and make the world a better place.

(B) Conclusions

The purpose of this research is to identify and analyze the elements that influence the decision-making process of Indian consumers regarding the utilization of drone delivery services (DDS). Combining the TAM and UTAUT2 models with privacy risk components allowed us to construct a comprehensive model. We reviewed the previous studies before carrying it out. In this study, we offered ten practical exploratory approaches to analyze the gap between customers' intentions and actual use of DFD services. To examine the data collected from 691 people all throughout India, we used statistical programs like SPSS and PLS-SEM. With these

programs, we were able to examine the data according to many criteria, including age, education level, location, and more. Privacy worries, performance problems, or delivery concerns are not determinants of DDS's acceptability. The factors that were included in this study explain 32.9% of the variance in DDS service use. But there are constraints that must be carefully considered in every subsequent works. Everyone involved in this study had zero background with DDS, and that is a crucial point to remember. After using drone delivery services, their opinions may alter. The study also has the drawback of focusing just on client impressions. As a result, future research should include online food purchases as a participant to obtain additional information regarding the stakeholders of the company. When it comes to any future research, I strongly advocate conducting a qualitative study. Not only do drone delivery services have the ability to transform corporate enterprises, but they also have the potential to revolutionise educational institutions, government agencies, and social organisations in general. Not only do they make it possible for academic institutions to explore new avenues of research and education, but they also give governments the capacity to enhance public services and infrastructure. These chances give firms the possibility to expand and flourish in their respective industries.

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